

M06 PROTECTED PRACTICE REVIEW

Open only after a genuine saved PBIX attempt

Compare logic/context -> repair -> close -> changed retry -> append evidence.

PBD1 review - Measures first: the calculation layer

1. Strong attempt should show

Base measures reconcile to protected practice anchors; implicit raw-column aggregation is not used as the final KPI logic.

2. Expected reasoning

Core teaching: Measures are model calculations evaluated when a visual/query asks for them. Calculated columns are row-by-row stored values. Analyst KPIs should normally be explicit measures because business logic stays named, reusable and testable. Use SUM for an existing additive column, SUMX when an expression must be evaluated row by row, DISTINCTCOUNT for business entities, COUNTROWS for rows, and DIVIDE for safe ratios. Guided pattern: Create explicit measures: Line Count, Order Count, Customer Count, Total Quantity, Gross Sales, Net Sales, Total Cost, Profit, Margin %, Average Order Value. Save a card/table page and explain SUM vs SUMX and DIVIDE.

3. What to compare

Compare the business definition, model prerequisite, active filter context, measure formula/branch and independent validation before comparing screenshots. Equivalent correct DAX is acceptable.

4. Likely repair

If totals look wrong, check the source column, aggregation choice and whether the KPI needs SUM or SUMX. If a ratio divides by zero, use DIVIDE rather than hiding the issue.

5. Fresh retry

Close this review. Perform the changed transfer task in a different context/page/measure as assigned. Save a new PBIX/evidence row; only the changed successful retry upgrades mastery.

PBD2 review - Evaluation context: why the same measure changes

1. Strong attempt should show

Matrix/slicer results match predictions or corrections; learner can name the active filters that explain each change.

2. Expected reasoning

Core teaching: Filter context is the set of filters active when a measure is evaluated: visual rows/columns, slicers, page/report filters, relationship propagation and DAX filters. The measure formula can stay identical while the result changes because the question/context changed. When a number surprises you, ask: what filters are active and which relationships carry them to the fact table? Guided pattern: Create Category x Region matrix with Net Sales and Profit plus Region/Category slicers. Predict three selections before clicking, test them, and write the observed contexts.

3. What to compare

Compare the business definition, model prerequisite, active filter context, measure formula/branch and independent validation before comparing screenshots. Equivalent correct DAX is acceptable.

4. Likely repair

If a number changes unexpectedly, remove slicers one at a time, inspect visual/page/report filters, then trace relationship filter flow to the fact table.

5. Fresh retry

Close this review. Perform the changed transfer task in a different context/page/measure as assigned. Save a new PBIX/evidence row; only the changed successful retry upgrades mastery.

PBD3 review - CALCULATE: deliberately changing context

1. Strong attempt should show

Share denominator behavior is correct; shares reconcile sensibly and do not all show 100%.

2. Expected reasoning

Core teaching: CALCULATE evaluates a measure/expression under modified filter context. Start from trusted base measures and change only the context needed for the question. REMOVEFILTERS is useful when a denominator must ignore a breakdown filter. A common bug is leaving the denominator filtered so every row appears to be 100%. Guided pattern: Create East Sales, Electronics Sales, All Category Sales and Category Share %. Validate category shares under a Region slicer and explain the denominator context.

3. What to compare

Compare the business definition, model prerequisite, active filter context, measure formula/branch and independent validation before comparing screenshots. Equivalent correct DAX is acceptable.

4. Likely repair

If each share row shows 100%, inspect the denominator context and REMOVEFILTERS target. If CALCULATE produces blank, check whether the filter value exists and whether a relationship can propagate it.

5. Fresh retry

Close this review. Perform the changed transfer task in a different context/page/measure as assigned. Save a new PBIX/evidence row; only the changed successful retry upgrades mastery.

PBD4 review - KPI measures and targets

1. Strong attempt should show

Target totals are evaluated at Region + Month grain; actual/target/variance/attainment reconcile to the target table and no unsupported daily target is presented.

2. Expected reasoning

Core teaching: Targets are facts at a defined grain. If a target table is Region + Month, the model must compare actual vs target at a compatible Region + Month grain unless the business supplies an allocation rule. $\text{Variance} = \text{actual} - \text{target}$; $\text{attainment} = \text{actual} / \text{target}$. A polished card at the wrong grain is still wrong. Guided pattern: Relate Regions and Date to Monthly_Targets; create Sales Target, Variance to Target and Target Attainment %. Compare by Region + Month and explain why daily target is unsupported.

3. What to compare

Compare the business definition, model prerequisite, active filter context, measure formula/branch and independent validation before comparing screenshots. Equivalent correct DAX is acceptable.

4. Likely repair

If target repeats or multiplies, stop and restate target grain. Check target-table key uniqueness and the Date/Region relationships before changing DAX.

5. Fresh retry

Close this review. Perform the changed transfer task in a different context/page/measure as assigned. Save a new PBIX/evidence row; only the changed successful retry upgrades mastery.

PBD5 review - Time intelligence that you can defend

1. Strong attempt should show

January 2026 previous month equals December 2025; prior-year measure uses the same comparable period; YTD increases logically across months.

2. Expected reasoning

Core teaching: Time intelligence depends on a proper Date table: one row per date, coverage across the required analysis period and an active relationship to the fact date. YTD, prior-year and prior-month measures are context transformations, not hard-coded lookups. January is a key validation period because previous month crosses a year boundary. Guided pattern: Create Sales YTD, Sales PY, YoY %, Previous Month Sales and MoM %. Build a monthly 2025-2026 line chart and test January 2026 against December 2025 plus one prior-year period.

3. What to compare

Compare the business definition, model prerequisite, active filter context, measure formula/branch and independent validation before comparing screenshots. Equivalent correct DAX is acceptable.

4. Likely repair

If previous month/YTD is blank or strange, validate the Date table coverage/continuity, the active Date relationship and the current date context before rewriting the measure.

5. Fresh retry

Close this review. Perform the changed transfer task in a different context/page/measure as assigned. Save a new PBIX/evidence row; only the changed successful retry upgrades mastery.

PBD6 review - Visual selection and page hierarchy

1. Strong attempt should show

Each visual answers a named question, formats are consistent, reading order is obvious and the page does not rely on color alone.

2. Expected reasoning

Core teaching: Visual choice is part of analysis. Cards communicate a few headline KPIs; line charts show ordered time; bars compare/rank categories; matrices support detail; scatterplots show numeric relationships. Page hierarchy should make the main question obvious before supporting detail. Avoid decoration that carries no information and do not rely on color alone. Guided pattern: Build a 16:9 Executive Summary: 4-5 KPI cards, monthly trend, Category bar, Region bar and one supporting matrix/table. Save a three-sentence design rationale.

3. What to compare

Compare the business definition, model prerequisite, active filter context, measure formula/branch and independent validation before comparing screenshots. Equivalent correct DAX is acceptable.

4. Likely repair

If the page feels crowded, remove visuals that duplicate another question, reduce slicers and keep 3-5 headline KPIs. If interpretation depends on color alone, add labels/signs.

5. Fresh retry

Close this review. Perform the changed transfer task in a different context/page/measure as assigned. Save a new PBIX/evidence row; only the changed successful retry upgrades mastery.

PBD7 review - Interactivity without confusion

1. Strong attempt should show

Filter/highlight/none behavior matches intent; drillthrough receives the selected context; tooltip values do not contradict the source visual.

2. Expected reasoning

Core teaching: Interactivity must reduce effort, not hide context. Use only high-value slicers. Explicitly inspect Edit interactions. Drillthrough is for a selected entity/detail path; report-page tooltips give compact supporting context. A visual interaction is correct only when its behavior matches the business question and validation totals remain consistent. Guided pattern: Add only useful Date/Region/Category slicers, inspect Edit interactions, create one drillthrough page and one report-page tooltip, then save one verified interaction state.

3. What to compare

Compare the business definition, model prerequisite, active filter context, measure formula/branch and independent validation before comparing screenshots. Equivalent correct DAX is acceptable.

4. Likely repair

If a slicer affects the wrong visual, use Edit interactions and test filter/highlight/none explicitly. If drillthrough is unavailable, verify the drillthrough field and selected context.

5. Fresh retry

Close this review. Perform the changed transfer task in a different context/page/measure as assigned. Save a new PBIX/evidence row; only the changed successful retry upgrades mastery.

PBD8 review - Report validation: prove the story is true

1. Strong attempt should show

At least seven validation checks have Expected/Actual/Status and include base, breakdown, target, time and interaction evidence.

2. Expected reasoning

Core teaching: Validation is a deliverable. Reconcile source rows, base measures, dimensional totals, target/time calculations, interactions and cross-page consistency. Test both grand total and breakdown grain because context bugs can hide at one level. Keep a QA page/log with Expected, Actual, Status and the evidence used. Guided pattern: Create a QA page/log and reconcile line count, order count, customer count, Net Sales and Profit. Add one target check, one time-intelligence check and one slicer-interaction check.

3. What to compare

Compare the business definition, model prerequisite, active filter context, measure formula/branch and independent validation before comparing screenshots. Equivalent correct DAX is acceptable.

4. Likely repair

If a grand total matches but categories do not, validate at the breakdown grain. If Expected vs Actual differs, isolate one measure/context at a time rather than rebuilding the report.

5. Fresh retry

Close this review. Perform the changed transfer task in a different context/page/measure as assigned. Save a new PBIX/evidence row; only the changed successful retry upgrades mastery.

PBD9 review - Publish, workspaces, refresh and sharing basics

1. Strong attempt should show

Workflow distinguishes publishing from data refresh and states account/tenant limitations accurately; no fabricated service evidence.

2. Expected reasoning

Core teaching: Publishing sends the Desktop report/semantic model to a Power BI workspace, subject to account and tenant permissions. Browser visual refresh is not the same as imported-data refresh. Scheduled/data refresh updates the semantic model from sources and may require credentials/gateway depending on source location. The course never requires buying a license. Guided pattern: If allowed, publish to My workspace and inspect report/semantic model; otherwise demonstrate Desktop Refresh and write the publish/refresh workflow accurately.

3. What to compare

Compare the business definition, model prerequisite, active filter context, measure formula/branch and independent validation before comparing screenshots. Equivalent correct DAX is acceptable.

4. Likely repair

If publish/refresh is unavailable, do not invent service success. Demonstrate Desktop refresh and write the correct service workflow/licensing caveat.

5. Fresh retry

Close this review. Perform the changed transfer task in a different context/page/measure as assigned. Save a new PBIX/evidence row; only the changed successful retry upgrades mastery.

PBD10 review - RLS basics, handoff and defense

1. Strong attempt should show

RLS role visibly restricts rows under View as roles; handoff includes KPI definitions, refresh/source notes, validation and security assumptions.

2. Expected reasoning

Core teaching: RLS restricts rows a user can query through model roles; a slicer or hidden page is not security. For beginner practice, a static role such as `Regions[Region] = "East"` is enough. A professional handoff also includes KPI definitions, PBIX/version, sources/refresh notes, validation results, RLS assumptions, known limitations and portfolio screenshots. Guided pattern: Create static East Only role, use View as roles, prove other regions disappear, then write the final handoff checklist and a 90-second project defense.

3. What to compare

Compare the business definition, model prerequisite, active filter context, measure formula/branch and independent validation before comparing screenshots. Equivalent correct DAX is acceptable.

4. Likely repair

If RLS test does not restrict rows, verify the role expression and relationship path, then use View as roles. Do not treat hidden pages or slicers as security.

5. Fresh retry

Close this review. Perform the changed transfer task in a different context/page/measure as assigned. Save a new PBIX/evidence row; only the changed successful retry upgrades mastery.